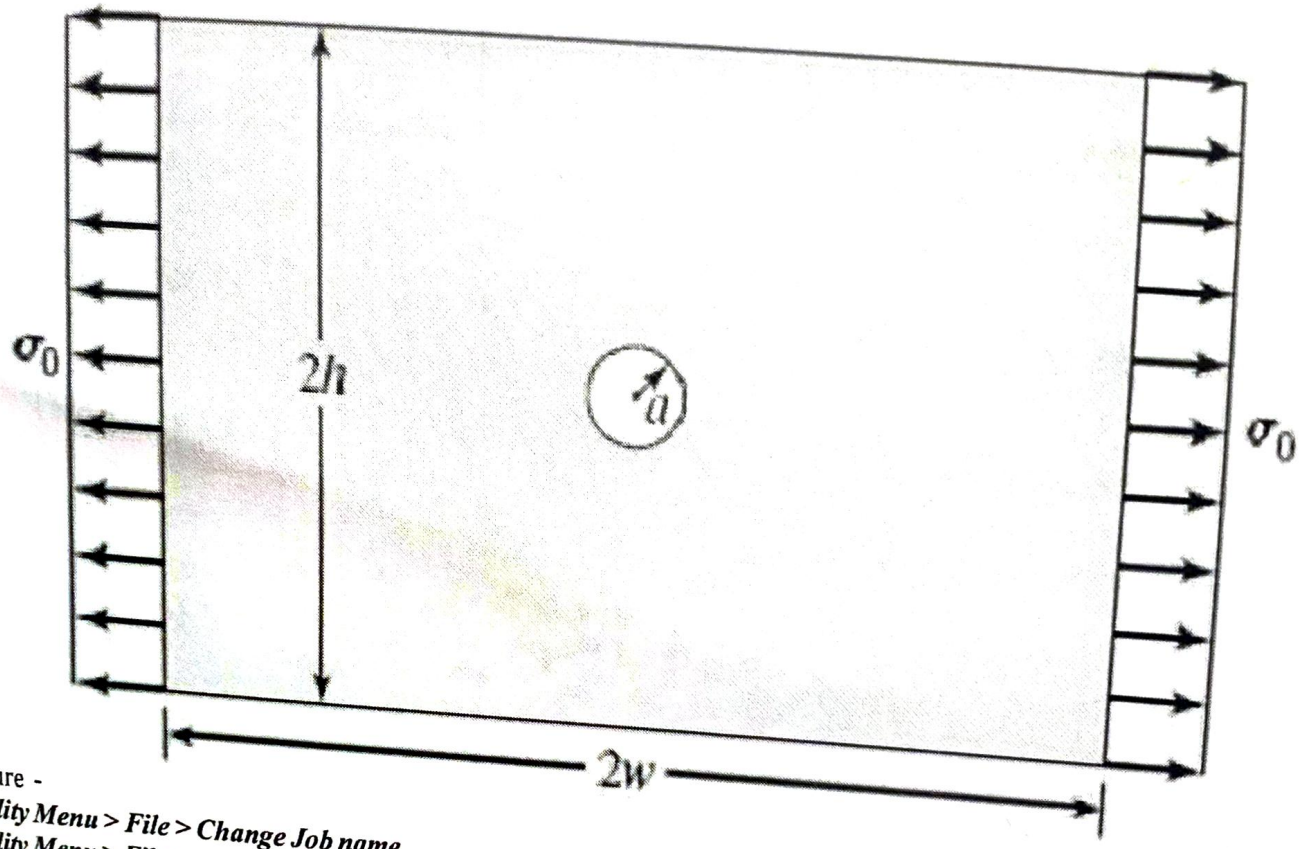
CHAPTER 4

ANSYS EXERCISE (PART – B)

PART B.1: STRESS ANALYSIS OF RECTANGULAR PLATE WITH A CIRCULAR HOLE

Problem 35

Stress analysis of a rectangular Plate with Hole. Let 2w = 100mm, 2h =50mm,a= 10mm, $\sigma_0 = 100$ Mpa.



Procedure -

1. *Utility Menu > File > Change Job name*
2. *Utility Menu > File > Change Title* > Stress analysis of a Plate
3. *Preprocessor > Element type > Add/Edit/Delete > Add > Solid > Quad 4 node 42*
4. *Preprocessor>Material Property> Young's Modulus > Enter Ex = 2e5 and Poisson's ratio = 0.3.*
5. *The model is symmetric we are going to create ¼ of the model.*
   Preprocessor>modeling >create> Area > Rectangle> By Dimensions> 0, 100 (X1, X2) (0, 50) for (Y1, Y2).
6. *Preprocessor>Modeling > Create> Area >Circle > Solid circle > (definedimensions). Enter WP X = 50, WP Y = 25, Radius = 10.*
6. *Preprocessor>modeling>operate>Booleans>subtract>areas> (select the bigger Area first click apply and select the smaller one and click OK).*
8. *Preprocessor >Meshing>Mesh tool>Smart size ON>Free> Quad >Mesh > Pick all.*
9. *Preprocessor>Load>apply>Structural>Displacement>symmetry B.C>on line (Select the line on X coordinates Zero and Y coordinates Zero and apply Symmetric BCs).*
10. *Preprocessor>Load>apply>Structural> pressure> on line > Enter VALI = 100.*
11. *Solution > Solve > Current LS*

12. *General postprocessor > Plot Result > Deform shape*
13. *General postprocessor> Plot Result> Contour Plot> Nodal Solution>Displacement Vector Sum.*
14. *General postprocessor > Plot Result > Contour Plot > Nodal Solution >Stress >von Misses stress*
15. *To Expand the Model.*

Plot control > Style > Symmetry Expansion > Periodic /Cyclic Symmetry>1/4 Expansion > OK.

Results:

	ANSYS	Theory
Deformation		
Stress		

Preprocessor > Modelling > Create > Area > Rectangle.
> By dimensions >, 0, 100 (x_1, x_2) & (10, 50) for (½)

Preprocessor > Modelling > Create > Area > Circle.
Solid Circle > Define Dimensions > Radius = 15, x = 75,
y = 40.

Preprocessor > Meshing > MeshTool > Smart size > On >
Set > Quad Mesh

Preprocessor > Load > Apply > Structural > Displacement

Preprocessor > Load > Apply > Structural > [illegible]